

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

January 2026

ALGEBRA (Ph.D. Version)

Instructions to the student

- Answer all six questions; each will be assigned a grade from 0 to 10.
 - Use a different booklet for each question. Write the problem number and your **code number (not your name)** on the outside of the booklet.
 - Keep scratch work on separate pages in the same booklet.
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- Let n be a positive odd integer and let D be the dihedral group of order $2n$. So G is generated by a, b with relations $a^2 = 1$, $b^n = 1$, $aba^{-1} = b^{-1}$. Let C be a cyclic subgroup of D .
 - Show that if C has even order then the order of C is 2.
 - Show that if C has odd order then C is a normal subgroup of D .
- Let V be a finite-dimensional vector space of dimension n over a field F . Let $T : V \rightarrow V$ be a linear transformation and let $f(X)$ be the characteristic polynomial of T .
 - Suppose $f(X)$ is irreducible. Let $0 \neq v \in V$. Show that $\{v, Tv, T^2v, \dots, T^{n-1}v\}$ is a basis of V . (*Hint:* The gcd of two polynomials can be written as a linear combination of the two polynomials.)
 - Suppose that $\{v, Tv, T^2v, \dots, T^{n-1}v\}$ is a basis of V for each nonzero $v \in V$. Show that $f(X)$ is irreducible.

(*Note:* You may not quote results about cyclic vectors since these results are what you are proving.)
- Let F be a field. Suppose that the additive group of F is a finitely generated abelian group. Show that F is a finite field.
- Let R be a commutative ring with 1. Let M be an R -module and let $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$ be an exact sequence of R -modules. It is known that

$$A \otimes M \rightarrow B \otimes M \rightarrow C \otimes M \rightarrow 0$$

is always exact (where $\otimes$ denotes $\otimes_R$). If $A \otimes M \rightarrow B \otimes M$ is an injection for all such exact sequences $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$, then M is said to be *flat*. Suppose that $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M \rightarrow 0$ is an exact sequence of R -modules. Let N be an R -module. This problem will show that if M is flat then

$$0 \rightarrow M_1 \otimes N \rightarrow M_2 \otimes N \rightarrow M \otimes N \rightarrow 0$$

is exact. This only requires showing that $M_1 \otimes N \rightarrow M_2 \otimes N$ is an injection.

- (a) Show that a free module is flat.
- (b) Write $N = F/K$, where F is free. Use the diagram

$$\begin{array}{ccccccc}
 & & & & 0 & & \\
 & & & & \downarrow & & \\
 & M_1 \otimes K & \xrightarrow{f_1} & M_2 \otimes K & \xrightarrow{g_1} & M \otimes K & \longrightarrow 0 \\
 & \downarrow v_1 & & \downarrow v_2 & & \downarrow v_3 & \\
 0 \longrightarrow & M_1 \otimes F & \xrightarrow{f_2} & M_2 \otimes F & \xrightarrow{g_2} & M \otimes F & \longrightarrow 0 \\
 & \downarrow w_1 & & \downarrow w_2 & & \downarrow w_3 & \\
 & M_1 \otimes N & \xrightarrow{f_3} & M_2 \otimes N & \xrightarrow{g_3} & M \otimes N & \longrightarrow 0 \\
 & \downarrow & & \downarrow & & \downarrow & \\
 & 0 & & 0 & & 0 &
 \end{array}$$

to show that $M_1 \otimes N \rightarrow M_2 \otimes N$ is an injection when M is flat.

5. Let $f(X) \in \mathbb{Q}[X]$ be irreducible of degree 4, and let r_1, r_2, r_3, r_4 be its roots in $\mathbb{C}$. Let L be the splitting field of $f(X)$. Assume $[L : \mathbb{Q}] = 24$.
- (a) (3 points) Suppose $i \neq j$ and $k \neq \ell$. Show that $r_i + r_j = r_k + r_\ell$ if and only if $\{i, j\} = \{k, \ell\}$.
- (b) (7 points) Show that $[\mathbb{Q}(r_1 + r_2) : \mathbb{Q}] = 6$.
6. Let G be a finite group, let $\rho : G \rightarrow GL(V)$ be a finite-dimensional representation of G on some complex vector space V , and let $\chi = \text{Trace}(\rho)$ be the character of ρ .
- (a) Let $g \in G$. Show that the eigenvalues of $\rho(g)$ are roots of unity and that $\chi(g^{-1}) = \overline{\chi(g)}$ (= the complex conjugate of $\chi(g)$).
- (b) Suppose that G has odd order, $\chi(g) \in \mathbb{Z}$ for all $g \in G$, and V has odd dimension. Show that

$$\sum_{g \in G} \chi(g)$$

is odd (and therefore nonzero). (*Hint:* $g \neq g^{-1}$ if $g \neq 1$.)

- (c) Suppose that G has odd order, $\chi(g) \in \mathbb{Z}$ for all $g \in G$, and V has odd dimension. Show that there is a nonzero vector $v \in V$ such that $\rho(g)(v) = v$ for all $g \in G$.